

SentinelX

4. Responsible AI Guardrail

Guardrail 1 — Automation Bias

One major risk identified during the design of SentinelX was the possibility of over-reliance on AI-generated outputs during sensitive intervention workflows. In high-stakes safety environments, analysts may unintentionally begin treating AI flags or evidence analyses as definitive truth rather than contextual support. This creates risks such as false escalation, contextual misinterpretation, and overconfidence in algorithmic outputs.

To reduce this risk, both AI components only ever produce explainable, attributable outputs — never opaque final classifications — and neither can take action on its own. The Bi-LSTM only flags; Gemini only describes what evidence shows. Analysts remain the final decision-makers in every case.

Guardrail 2 — Privacy and Data Minimization in Detect Threat

The second major risk is specific to the Detect Threat feature: continuous audio capture is inherently privacy-sensitive, and a poorly-designed implementation could create exactly the kind of surveillance tool the challenge brief explicitly warns against.

To reduce this risk, SentinelX was designed around strict data minimization: raw audio is transcribed to text on the device and discarded immediately — it is never stored, never uploaded, and never seen by any server or third party. Only the resulting text classification (threat or non-threat) and a transcript snippet ever leave the device, and only when a threat is actually detected. The Bi-LSTM model itself runs entirely on-device, so no audio or transcript is required to leave the phone for inference to happen. Detect Threat is also strictly opt-in — it only operates when the user has deliberately toggled it on, never silently or by default.

Other safeguards built into the system include:

- mandatory human-triggered evidence analysis for Safe Chat reports,
- no autonomous intervention escalation in any flow,

- structured, consent-based intake rather than passive background monitoring,
- explainable, attributable outputs instead of opaque blended classifications,
- transparent analyst review interfaces,
- and explicit organizational accountability boundaries.

Human analysts remain the final decision-makers throughout the entire intervention process. SentinelX was designed to ensure that AI remains an assistive, narrowly-scoped intelligence layer rather than an autonomous authority system.

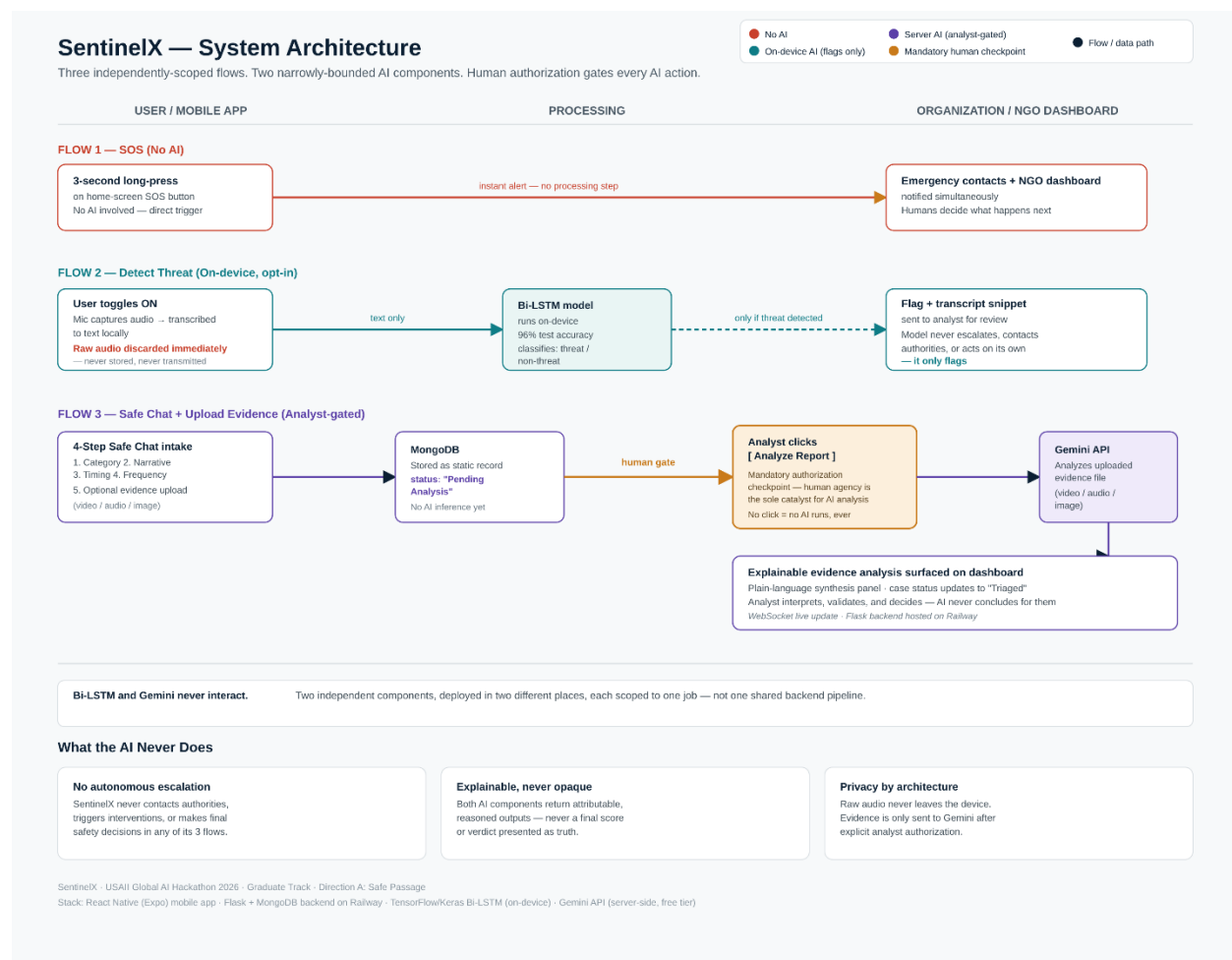


Fig 2 SentinelX System Architecture.